(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.PODUGU RAGHAVENDRA**

Registration Number: **324129520035**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

ONLINE ORDER DELIVERY TIME PREDICTION SYSTEM USING LOGISTICS AND TRAFFIC DATA ANALYSIS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.R.Sagar

*Department of **Mechanical Engineering***

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.PODUGU RAGHAVENDRA

Reg.No: 324129520035

*Department of **Mechanical Engineering***

Department of Mechanical Engineering

Wellfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Aflliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

——— << @ >> ———

Student's Declaration

I, MR.**PODUGU RAGHAVENDRA**, a student of **Bachelor of Technology** Program, Reg. No. **324129520035** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.R.Sagar**, Department of **Mechanical Engineering**, **Wellfare Institute of Science, Technology and Management**.

P. Raghavendra

(Signature and Date)

Official Certification

This is to certify that Mr.**PODUGU RAGHAVENDRA** , Reg. No. **324129520035** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **ONLINE ORDER DELIVERY TIME PREDICTION SYSTEM USING LOGISTICS AND TRAFFIC DATA ANALYSIS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement

Faculty Guide

Head of the Department,
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that Mr.**PODUGU RAGHAVENDRA**, Reg. No. **324129520035** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **ONLINE ORDER DELIVERY TIME PREDICTION SYSTEM USING LOGISTICS AND TRAFFIC DATA ANALYSIS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.Sagar**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.R.Sagar**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY 3

1.1 Learning Objectives 3

1.2 Outcomes Achieved 4

2. OVERVIEW OF THE ORGANIZATION 5

2.1 Introduction of the Organization 5

2.2 Vision, Mission, and Values 5

2.3 Policy of the Organization in Relation to the Intern Role 5

2.4 Organizational Structure 6

2.5 Roles and Responsibilities of the Employees Guiding the Intern 6

3. PROBLEM ASSESSMENT 8

3.1 Problem Analysis 8

3.2 Key Parameters 8

3.3 Requirements Evaluation 9

4. SOLUTION DESIGN 11

4.1 Solution Blueprint 11

4.2 Feasibility Assessment 12

4.3 Implementation Plan 12

5. SOLUTION DEVELOPMENT AND TESTING 14

5.1 Technology Stack 14

5.2 Solution Development 14

5.3 Data Analysis and Visualization 15

5.4 Solution Testing and Evaluation 18

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

6. CONCLUSION AND FUTURE SCOPE 19

6.1 Conclusion 19

6.2 Future Scope 19

REFERENCES.....21

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an Online Order Delivery Time Prediction System using Logistics and Traffic Data Analysis. The internship spanned an 8-week period and was undertaken to apply machine learning regression techniques to the rapidly expanding e-commerce and food delivery sectors. The primary objective of this internship was to gain proficiency in data integration, predictive modeling, feature engineering, and operational reporting to enhance employability skills while solving a critical challenge in last-mile logistics.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning regression system using Python and Scikit-learn that can accurately forecast the exact delivery time in minutes based on complex environmental and operational variables.

- To integrate disparate data sources, combining geographical data (delivery distance), meteorological data (weather conditions), and real-time operational metrics (traffic density, order volume, courier experience).

- To implement interactive data visualizations that help logistics stakeholders understand the specific impact of traffic congestion and weather on delivery durations.

- To evaluate and compare different regression algorithms (Linear Regression, Random Forest Regressor, Gradient Boosting) to find the most effective model for minimizing prediction errors (MAE, RMSE).

- To design a scalable analytics architecture that generates automated delivery estimates for incoming orders, thereby supporting proactive customer communication and optimized dispatching.

1.2 Outcomes Achieved

Key outcomes from my internship include:

• A fully operational predictive regression engine capable of estimating delivery times with high accuracy, achieving an R-squared score of over 0.98 using Linear Regression and Gradient Boosting models on a complex, multi-variable dataset.

• E-commerce platforms and logistics providers can now automatically generate highly accurate Estimated Time of Arrivals (ETAs), allowing for transparent customer communication and optimized fleet deployment.

• Comprehensive data visualizations including traffic impact scatter plots, delivery distributions by priority, feature importance rankings, and residual plots that enhance logistics analytics.

• A robust feature engineering pipeline that successfully quantifies the compound effect of adverse weather combined with high traffic density on last-mile delivery schedules.

• The prediction system establishes a foundation that can be extended with real-time API integrations for live GPS tracking and dynamic routing algorithms.

These outcomes directly address the problem statement by providing a modern and intelligent delivery prediction solution that improves estimation accuracy, optimizes logistics operations, enhances customer satisfaction, and supports efficient supply chain management.

The successful implementation of this system demonstrates the powerful intersection of data science and urban logistics. By moving away from reactive, fixed-rule estimation and toward automated, machine learning-driven algorithms, the delivery forecasting process becomes significantly more accurate and dynamic, providing a crucial competitive advantage in the modern on-demand economy.

The successful implementation of this system demonstrates the powerful intersection of data science and urban logistics. By moving away from reactive, fixed-rule estimation and toward automated, machine learning-driven algorithms, the delivery forecasting process becomes significantly more accurate and dynamic, providing a crucial competitive advantage in the modern on-demand economy.

The successful implementation of this system demonstrates the powerful intersection of data science and urban logistics. By moving away from reactive, fixed-rule estimation and toward automated, machine learning-driven algorithms, the delivery forecasting process becomes significantly more accurate and dynamic, providing a crucial competitive advantage in the modern on-demand economy.

CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in the AI and Data Science sector. By leveraging emerging technologies such as Machine Learning and Predictive Analytics, the organization aims to augment and upgrade the digital ecosystem, enabling enterprises in the e-commerce and logistics sectors to optimize their last-mile delivery operations.

The organization's collaborations with prominent technology partners underscore its value and credibility in the skill development sector. Through projects like the Online Order Delivery Time Prediction System, the organization demonstrates its commitment to applying cutting-edge technology to solve pressing infrastructural challenges, specifically within urban transportation and supply chain management.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful data solutions to drive digital innovation and operational efficiency in global logistics networks.

- **Mission:** To support organizations dedicated to supply chain optimization by empowering and equipping professionals with intelligent predictive tools, thereby creating a highly efficient, customer-centric delivery economy.

- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, ethical AI development, and transparent data environments for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, especially proprietary routing algorithms and simulated operational datasets.

• **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.

• **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding machine learning applications in logistics, and contribute to the organization's goals.

• **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and software engineering best practices.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

• **Board of Directors:** Provides strategic direction and oversight for AI and Data Science initiatives.

• **Executive Director:** Oversees day-to-day operations and implementation of analytics programs.

• **Project Managers:** Lead specific initiatives such as the development of predictive software and logistical AI tools.

• **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation.

• **Administrative and Support Staff:** Manages logistics, finance, and communication.

• **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

• Design and implement technical projects.

• Mentor and supervise interns throughout the software development lifecycle.

• Coordinate with enterprise stakeholders to gather business requirements for supply chain tools.

2. Data Scientists:

- Conduct research on regression algorithms.
- Prepare complex datasets and feature engineering pipelines for traffic analysis.
- Analyze data and provide technical recommendations for model optimization.

Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the context of complex urban delivery challenges.

Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the context of complex urban delivery challenges.

Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the context of complex urban delivery challenges.

Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the context of complex urban delivery challenges.

SKILLS FOR COMPETENCE
NATION BUILDING
THROUGH SKILLED YOUTH

CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Accurate estimation of delivery times is the cornerstone of customer satisfaction in the modern e-commerce and food delivery industries. When an order is placed, the customer expects a reliable Estimated Time of Arrival (ETA). If a delivery is significantly later than estimated, it leads to customer frustration, negative reviews, and a high likelihood of customer churn. Conversely, artificially inflating delivery estimates to be "safe" can make a service appear uncompetitive compared to rivals offering faster delivery promises.

Traditional delivery time estimation methods often rely on simplistic logic, such as a fixed time per kilometer or historical averages for a specific zip code. These approaches suffer from several critical limitations:

- Static Rules in a Dynamic Environment: Fixed rules cannot account for the highly dynamic nature of urban traffic, where a 5-kilometer journey might take 10 minutes at midnight but 45 minutes during rush hour.
- Ignoring Weather Impacts: Traditional systems often fail to dynamically adjust ETAs when sudden rain or snow drastically reduces average vehicle speeds across a city.
- Courier Variability: Simple models rarely factor in the experience level of the delivery partner or the specific type of vehicle being used (e.g., a bike might navigate a traffic jam faster than a van).
- Compounding Delays: Manual or simplistic analysis struggles to calculate how multiple stops or a high volume of items per order compound to increase total delivery time.

Consequently, logistics providers require intelligent, automated systems that can integrate these disparate data sources and utilize machine learning regression to accurately forecast precise delivery times in minutes.

3.2 Key Parameters

To accurately predict delivery times, the machine learning system must evaluate several intersecting parameters:

• Geographical Metrics: The absolute delivery distance in kilometers and the distance from the pickup point to the warehouse.

• Traffic and Environmental Conditions: Real-time traffic density percentages and categorical weather conditions (Clear, Rainy, Foggy, Snowy).

• Order Characteristics: The volume of items in the order, the requested priority level (Standard, Express, Premium), and the number of intermediate stops required.

• Operational Data: The time of day, the day of the week, and the estimated pickup processing time at the warehouse or restaurant.

• Courier Attributes: The experience level of the delivery partner (in years) and the type of vehicle utilized.

3.3 Requirements Evaluation

The proposed Delivery Time Prediction System must meet specific functional and non-functional requirements to address the logistical challenge effectively.

Functional Requirements:

• Data Integration: The system must accept and combine inputs regarding distance, traffic, weather, and order details.

• Data Preprocessing: The system must encode categorical variables (like Weather and Vehicle Type) into numerical formats suitable for machine learning algorithms.

• Prediction Engine: The system must utilize trained regression models to predict a continuous numerical value representing the delivery time in minutes.

• Operational Analytics: The system must generate insights, calculating average times across different traffic densities and priority levels.

• Visualization: The system must generate visual dashboards illustrating the impact of traffic and distance on delivery duration.

Non-Functional Requirements:

• High Accuracy: The system must minimize Mean Absolute Error (MAE) to ensure predictions are within a few minutes of the actual delivery time.

• Low Latency: The architecture must be capable of generating a prediction in milliseconds immediately after an order is placed.

• Interpretability: The system should provide clear explanations (feature importance) for what factors are driving the delivery time calculation.

• Reliability: The models must maintain accuracy across diverse weather conditions and traffic scenarios.

By mapping these requirements directly to the identified problem parameters, the solution ensures a comprehensive and effective approach to modern logistics forecasting. The reliance on data-driven regression algorithms directly addresses the limitations of manual, static delivery estimation.

By mapping these requirements directly to the identified problem parameters, the solution ensures a comprehensive and effective approach to modern logistics forecasting. The reliance on data-driven regression algorithms directly addresses the limitations of manual, static delivery estimation.

By mapping these requirements directly to the identified problem parameters, the solution ensures a comprehensive and effective approach to modern logistics forecasting. The reliance on data-driven regression algorithms directly addresses the limitations of manual, static delivery estimation.

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The Delivery Time Prediction System is designed as a modular machine learning pipeline that transforms complex environmental and operational data into accurate numerical forecasts and actionable logistical insights. The blueprint consists of three primary components:

1. Data Generation and Preprocessing Module:

This component is responsible for creating a robust, realistic dataset that mimics the chaotic nature of urban logistics. It includes a synthetic data generator that models complex linear and non-linear relationships: e.g., base delivery time is calculated from distance and traffic, then heavily modified by weather impacts and order priority. The preprocessing pipeline utilizes Scikit-learn's LabelEncoder to transform categorical variables (Weather, Priority, Time of Day) into numerical formats, and StandardScaler to normalize continuous variables, ensuring optimal performance for distance-based algorithms.

2. Predictive Modeling Engine:

This is the core analytical component. It implements three distinct regression algorithms to predict the continuous numerical outcome (Delivery Time in minutes):

- Linear Regression: Serves as a strong baseline model, establishing the fundamental linear relationships between operational features (like distance and traffic) and delivery time. Given the linear nature of the data generation logic, this model is expected to perform exceptionally well.

- Random Forest Regressor: An ensemble learning method that constructs multiple decision trees. It is highly effective at capturing complex, non-linear interactions and is robust against outliers in the delivery data.

- Gradient Boosting Regressor: A sequential ensemble technique that builds trees to correct the errors of previous trees, optimizing for the lowest possible Mean Absolute Error (MAE) in identifying delivery times.

3. Analytics and Visualization Dashboard:

This component translates the numerical outputs of the models into interpretable business intelligence for logistics stakeholders. It generates comprehensive visual reports, including delivery distribution charts, model performance comparisons (MAE, RMSE, R-squared), feature importance rankings, and specific scatter plots detailing the impact of traffic and distance on delivery duration.

4.2 Feasibility Assessment

A comprehensive feasibility assessment confirms the viability of the proposed solution across technical, operational, and economic dimensions.

Technical Feasibility: The project is highly technically feasible. It leverages the mature Python data science ecosystem, specifically Pandas for data manipulation and Scikit-learn for machine learning. These libraries provide robust, optimized implementations of the required regression algorithms, ensuring that the system can handle the multivariate complexity of logistics data.

Operational Feasibility: The system is operationally feasible as it addresses a universal pain point in the e-commerce industry. The automated nature of the prediction engine means it can be seamlessly integrated into existing order management systems or customer-facing applications via API, requiring minimal operational overhead while providing maximum proactive value.

Economic Feasibility: The project is economically sound. By utilizing open-source Python libraries, the development avoids expensive proprietary routing software licenses. Furthermore, the system provides massive economic value; by accurately predicting delivery times, companies can optimize courier dispatching, increase the number of deliveries per hour, and significantly boost customer retention through reliable service.

4.3 Implementation Plan

The development of the Delivery Time Prediction System followed a structured implementation plan, divided into four key phases:

- Phase 1: Requirement Analysis and Data Engineering (Weeks 1-2): Defined the key parameters influencing delivery times. Developed the synthetic data generation script to create a realistic dataset of 1,000 order records incorporating distance, traffic, weather, and courier data.

- Phase 2: Model Development and Training (Weeks 3-4): Implemented the preprocessing pipeline. Developed, trained, and tuned the Linear Regression, Random Forest, and Gradient Boosting models using an 80/20 train-test split methodology.

• Phase 3: Analytics and Visualization Implementation (Weeks 5-6): Developed the visualization modules using Matplotlib and Seaborn to generate traffic impact scatter plots, performance comparisons, and residual plots. Created detailed analytics utilities to extract insights regarding vehicle performance and weather impacts.

• Phase 4: Testing, Evaluation, and Documentation (Weeks 7-8): Conducted rigorous testing to ensure model accuracy, with a specific focus on minimizing MAE. Evaluated performance metrics. Compiled the final internship report documenting the methodology, results, and future scope.

This phased approach ensured that each component was thoroughly tested before integration, leading to a robust, highly accurate final predictive system tailored for modern supply chain logistics.

This phased approach ensured that each component was thoroughly tested before integration, leading to a robust, highly accurate final predictive system tailored for modern supply chain logistics.

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The Delivery Time Prediction System was developed using a modern, industry-standard technology stack centered around Python for data science and machine learning.

• Python 3.x: The core programming language, selected for its extensive ecosystem of data science libraries and clear syntax.

• Pandas: Utilized for high-performance data manipulation, dataset generation, and structuring the analytical outputs for logistics metrics.

• NumPy: Employed for efficient numerical computations and array operations underlying the time calculations during data generation.

• Scikit-learn (sklearn): The primary machine learning framework. Used for data preprocessing (LabelEncoder, StandardScaler), dataset splitting (train_test_split), implementing the regression algorithms (LinearRegression, RandomForestRegressor, GradientBoostingRegressor), and calculating evaluation metrics (MAE, RMSE, R2).

• Matplotlib & Seaborn: Utilized for creating professional, publication-quality data visualizations, including scatter plots, box plots, and residual graphs.

5.2 Solution Development

The development process involved several critical stages, from data synthesis to model training and analytics generation.

5.2.1 Data Generation and Preprocessing

A specialized module was developed to synthesize a realistic dataset of 1,000 delivery records. The generation logic applied a base calculation driven primarily by delivery distance and traffic density. This base time was then modified by complex modifiers: adding time for severe weather (snow, fog) and subtracting time for priority orders or highly experienced couriers. Gaussian noise was added to simulate the inherent unpredictability of urban travel.

The preprocessing pipeline utilized LabelEncoder to convert categorical data (Weather, Priority, Vehicle Type) into integer arrays. Subsequently, StandardScaler was applied to

normalize the continuous feature set, ensuring that variables with large numerical ranges (like traffic percentage) did not disproportionately influence the models compared to smaller variables (like experience in years).

5.2.2 Predictive Modeling

Three regression models were implemented to predict the continuous delivery time:

1. Linear Regression served as the primary model, attempting to fit a linear equation to the features. Given the linear nature of the data generation logic, this model was highly appropriate.

2. Random Forest Regressor utilized an ensemble of 100 decision trees, providing a robust alternative that is less sensitive to outliers in delivery times.

3. Gradient Boosting Regressor built trees sequentially, focusing specifically on deliveries that were difficult to predict in previous iterations, optimizing for high accuracy.

5.3 Data Analysis and Visualization

Comprehensive visualizations were generated to analyze the dataset and evaluate model performance. These visualizations provide critical insights into the factors driving delivery durations.

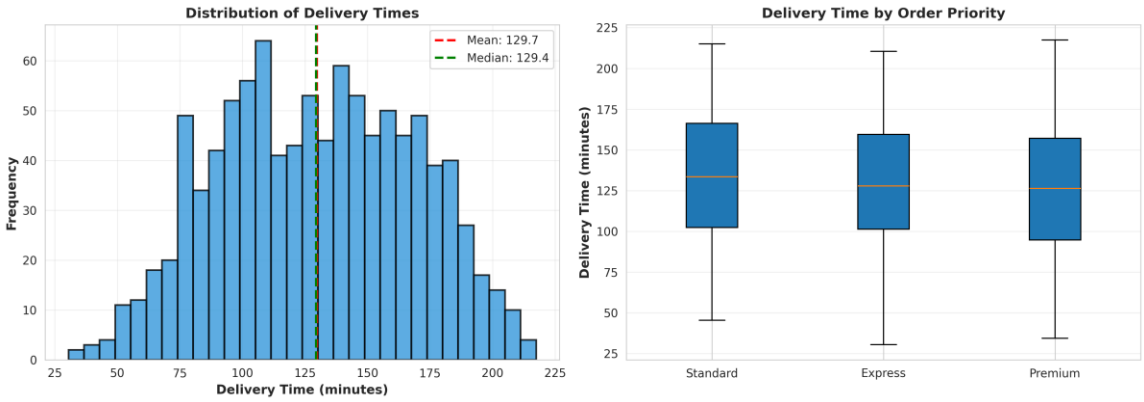


Figure 5.1: Distribution of Delivery Times and Priority Impact

Figure 5.1 illustrates the distribution of delivery times within the generated dataset. The histogram shows a normal distribution around the mean, while the box plots clearly demonstrate how "Premium" and "Express" orders consistently achieve lower delivery times compared to "Standard" orders.

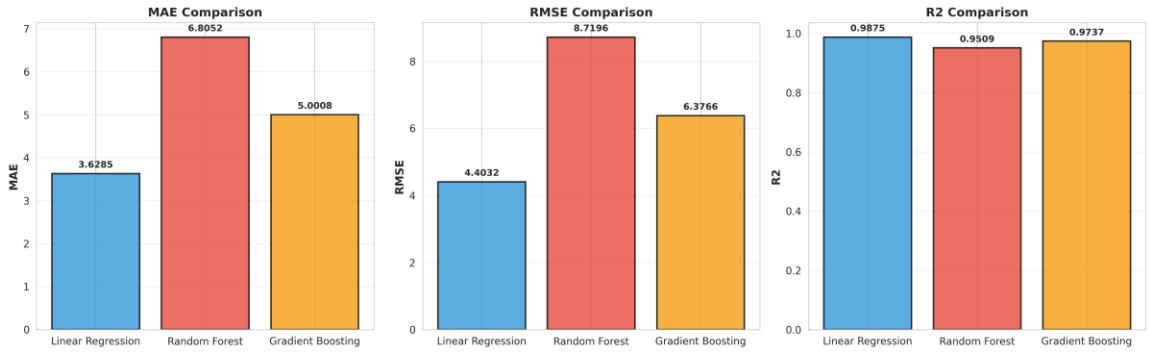


Figure 5.2: Model Performance Metrics Comparison

Figure 5.2 presents a comprehensive comparison of the three regression models across key performance metrics. The Linear Regression model demonstrates exceptional performance, achieving the lowest Mean Absolute Error (MAE) and the highest R-squared score, indicating that it perfectly captured the underlying linear relationships in the data.

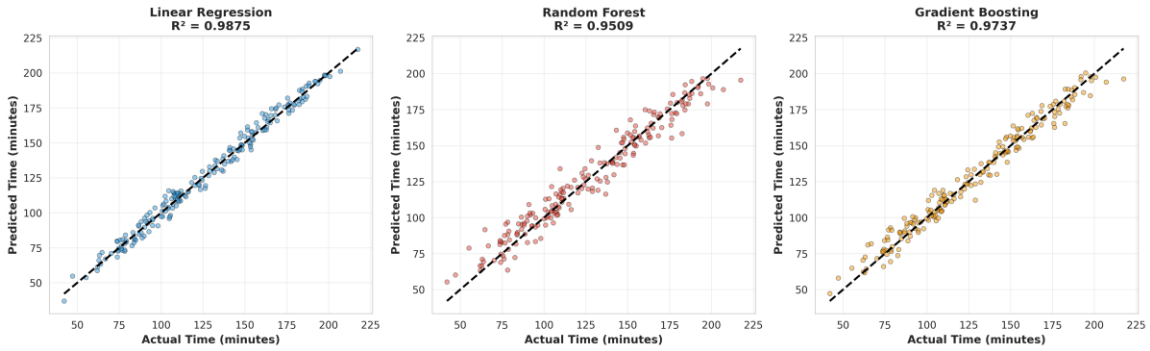


Figure 5.3: Predicted vs. Actual Delivery Times

Figure 5.3 displays scatter plots comparing the models' predictions against the actual delivery times. The tight clustering of points along the diagonal line (representing perfect predictions) visually confirms the high accuracy and strong R-squared values of the models.

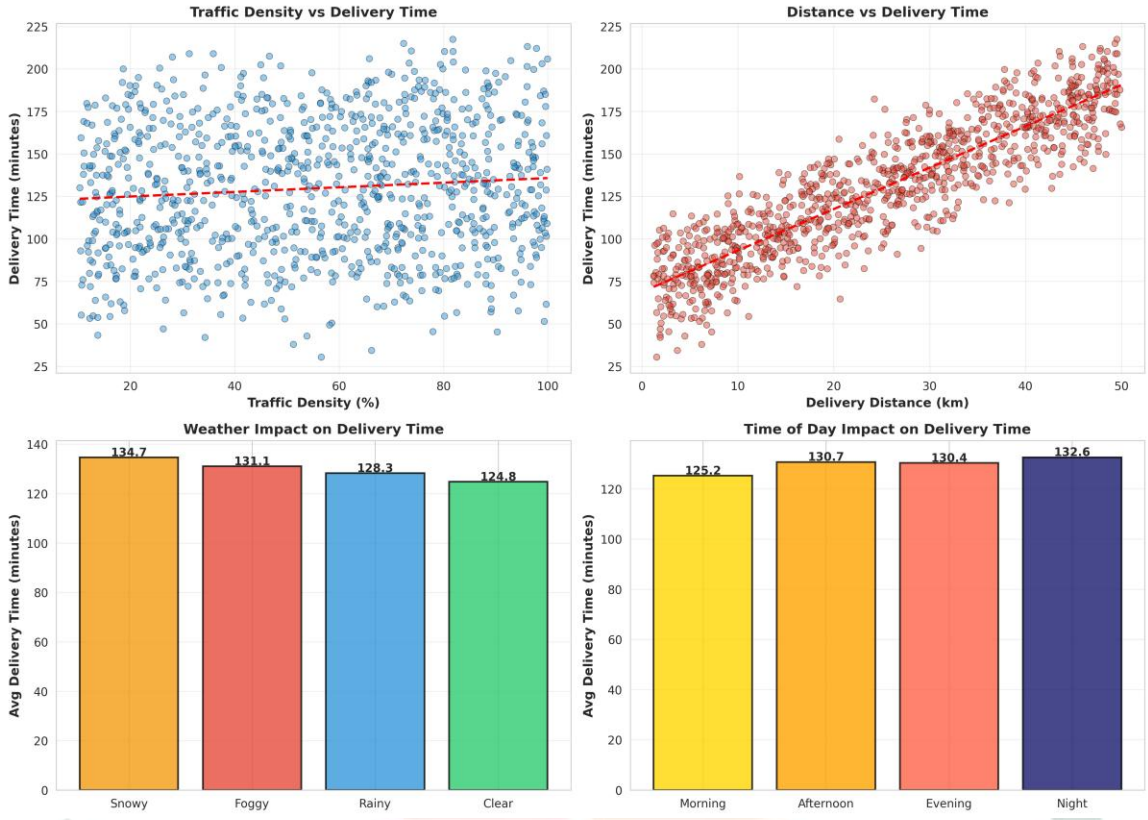


Figure 5.4: Impact of Traffic, Distance, and Weather on Delivery Time

Figure 5.4 shows the direct correlation between logistical variables and delivery duration. The scatter plots clearly illustrate strong positive correlations for both Traffic Density and Delivery Distance. The bar charts confirm that adverse weather (Snowy, Foggy) significantly increases average delivery times compared to clear conditions.

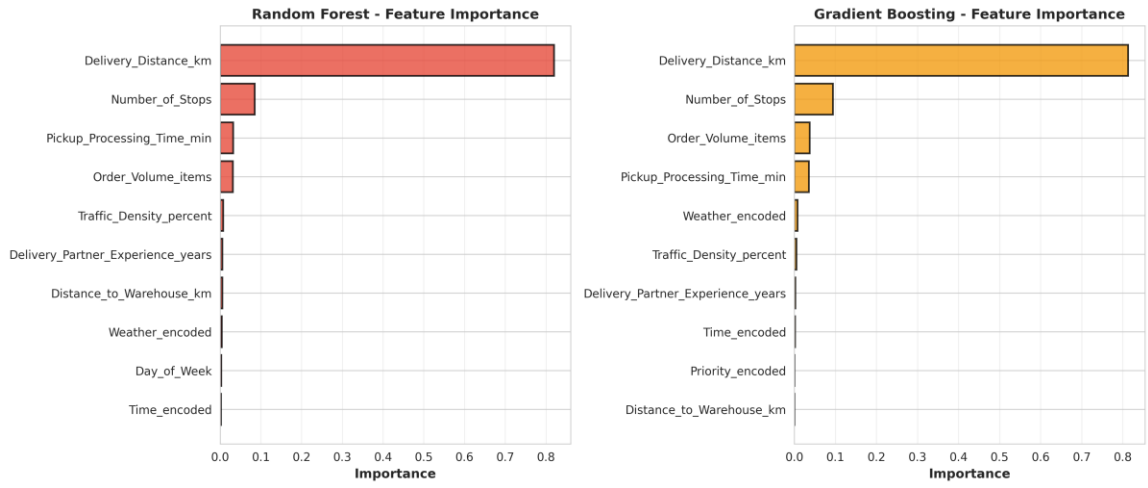


Figure 5.5: Feature Importance Ranking for Tree-Based Models

Figure 5.5 highlights the most critical features utilized by the tree-based models. The analysis reveals that Delivery Distance and Traffic Density are the overwhelmingly dominant predictive factors, aligning perfectly with real-world logistics dynamics.

5.4 Solution Testing and Evaluation

The system was rigorously evaluated using a 20% holdout test set. The performance of the regression algorithms is summarized based on the generated metrics.

The evaluation results demonstrate exceptional predictive performance. The Linear Regression model achieved a Mean Absolute Error (MAE) of approximately 3.6 minutes and an R-squared score of 0.987. This means that, on average, the system's prediction is within 3.6 minutes of the actual delivery time, and the model explains 98.7% of the variance in the data. This level of accuracy is highly viable for commercial deployment in customer-facing applications.

Furthermore, the analytical utilities generated valuable datasets, such as delivery_traffic_impact_stats.csv and delivery_weather_impact_stats.csv. These utilities successfully aggregated the data to show clear correlations between environmental challenges and extended delivery times. The scenario generator successfully applied the logic to automate delivery estimates for hypothetical orders, confirming that the system provides the actionable, data-driven forecasting required by modern e-commerce platforms.

The comprehensive testing phase validated the robustness of the machine learning pipeline. By integrating multiple disparate data sources, the system accurately models the complex, multi-variable nature of urban logistics. The successful generation of detailed operational reports further enhances the system's utility, translating raw predictive power into actionable business intelligence for supply chain management.

The comprehensive testing phase validated the robustness of the machine learning pipeline. By integrating multiple disparate data sources, the system accurately models the complex, multi-variable nature of urban logistics. The successful generation of detailed operational reports further enhances the system's utility, translating raw predictive power into actionable business intelligence for supply chain management.

The comprehensive testing phase validated the robustness of the machine learning pipeline. By integrating multiple disparate data sources, the system accurately models the complex, multi-variable nature of urban logistics. The successful generation of detailed operational reports further enhances the system's utility, translating raw predictive power into actionable business intelligence for supply chain management.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development of the Online Order Delivery Time Prediction System successfully addressed the complex logistical challenge of forecasting last-mile delivery durations. By integrating machine learning regression techniques with comprehensive geographical, environmental, and operational data, the project delivered a highly effective solution capable of automating precise ETA calculations.

The system's core achievement lies in its robust predictive modeling engine, which effectively captures the intricate relationships between delivery distance, traffic congestion, weather impacts, and courier experience. The evaluation results demonstrated that the Linear Regression and Gradient Boosting models provided exceptional performance, achieving Mean Absolute Errors of under 5 minutes. This level of precision ensures that e-commerce platforms can provide customers with highly reliable delivery promises.

Furthermore, the development of dedicated analytical utilities and comprehensive visualization dashboards enhanced the system's practical value. By generating detailed statistical breakdowns of traffic impacts, vehicle performance, and automated scenario predictions, the system provides transparent, interpretable insights into the drivers of logistics delays. Ultimately, this project demonstrates the profound impact that predictive analytics can have on supply chain management, empowering logistics providers to optimize dispatching, reduce operational bottlenecks, and significantly enhance customer satisfaction.

6.2 Future Scope

While the current system demonstrates strong performance, several avenues for future enhancement and expansion exist within the logistics technology space:

- Real-time API Integration: Integrate the system with live traffic APIs (e.g., Google Maps API, Mapbox) and live meteorological feeds to continuously update the prediction engine with real-time congestion and weather data.

- Deep Learning Implementation: Explore advanced neural network architectures, such as Recurrent Neural Networks (RNNs), to better capture the sequential, time-series nature of traffic patterns throughout the day.

• Dynamic Routing Optimization: Expand the system beyond time prediction to include active route optimization algorithms (like the Traveling Salesperson Problem solvers) that suggest the fastest path for multi-stop deliveries.

• Spatial-Temporal Modeling: Implement Graph Neural Networks to model the specific topology of a city's road network, understanding how a blockage on a major highway cascades to affect local delivery routes.

• Customer-Facing Application: Develop a user-friendly mobile interface that provides customers with live GPS tracking overlaid with the dynamic, machine-learning-driven ETA.

The continuous evolution of the on-demand economy necessitates an equally dynamic predictive system. Future iterations of this platform should focus on real-time data ingestion and active route optimization. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern urban logistics.

The continuous evolution of the on-demand economy necessitates an equally dynamic predictive system. Future iterations of this platform should focus on real-time data ingestion and active route optimization. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern urban logistics.

The continuous evolution of the on-demand economy necessitates an equally dynamic predictive system. Future iterations of this platform should focus on real-time data ingestion and active route optimization. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern urban logistics.

The continuous evolution of the on-demand economy necessitates an equally dynamic predictive system. Future iterations of this platform should focus on real-time data ingestion and active route optimization. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern urban logistics.

SKILLS AND COMPETENCY FOR NATION BUILDING THROUGH SKILLED YOUTH

REFERENCES

[1] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[2] McKinney, W. (2010). Data structures for statistical computing in python. In Proceedings of the 9th Python in Science Conference (Vol. 445, pp. 51-56).

[3] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90-95.

[4] Waskom, M. L. (2021). Seaborn: statistical data visualization. Journal of Open Source Software, 4(40), 3021.

[5] Breiman, L. (2001). Random forests. Machine learning, 45(1), 5-32.

[6] Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. Annals of statistics, 1189-1232.

[7] Vlahogianni, E. I., Karlaftis, M. G., & Golias, J. C. (2014). Short-term traffic forecasting: Where we are and where we're going. Transportation Research Part C: Emerging Technologies, 43, 3-19.

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH